

DE TOUR VILLAGE, MI, USA

WMO: 997257

Lat: 45.9925N

Lon: 83.8983W

Elev: 589

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.9	-3.5	N/A	N/A	N/A	N/A	N/A	N/A	33.4	22.6	26.6	22.7	8.7	340	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	79.1	N/A	76.0	N/A	73.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	9.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.4	22.7	19.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	16.9	16.9	25.9	38.4	49.6	60.0	66.2	65.1	58.5	47.2	35.6	24.7		
	DBStd	19.08	11.50	9.78	9.39	6.78	6.63	5.66	5.07	4.46	6.12	6.66	7.94	9.02		
	HDD50	4504	1025	928	748	350	89	3	0	0	7	136	434	784		
	HDD65	8484	1490	1348	1213	798	478	166	48	52	206	553	883	1249		
	CDD50	1667	0	0	0	2	78	304	501	470	263	48	1	0		
	CDD65	172	0	0	0	0	1	17	84	57	13	0	0	0		
	CDH74	486	0	0	0	0	8	60	277	127	14	0	0	0		
	CDH80	61	0	0	0	0	1	8	40	12	0	0	0	0		
	WSAvg	9.0	9.6	9.5	9.9	10.3	9.5	8.6	7.5	7.7	7.9	9.0	9.3	9.7		
	PrecAvg	31.6	2.1	1.3	1.7	2.3	2.8	3.2	3.3	3.2	3.5	3.4	2.6	2.2		
	PrecMax	40.2	4.8	2.7	4.5	4.3	5.2	6.7	7.3	6.0	7.1	6.7	6.1	4.6		
	PrecMin	21.8	0.6	0.4	0.1	0.6	0.5	1.0	1.3	0.6	0.9	1.2	0.7	0.5		
	PrecStd	4.9	1.0	0.7	1.1	1.0	1.1	1.5	1.5	1.5	1.7	1.4	1.2	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	39.4	51.1	61.1	74.6	80.7	84.4	81.6	76.0	65.9	54.3	43.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	38.3	35.5	44.0	54.3	67.8	74.7	80.3	77.4	72.2	61.9	51.0	40.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	35.9	33.4	40.4	50.5	63.4	71.1	77.0	74.7	69.3	59.2	48.2	38.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	33.2	31.6	37.3	47.4	59.7	68.2	74.1	72.1	67.3	56.8	45.9	36.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	13.2	13.6	11.9	13.3	12.8	12.6	10.9	10.6	9.0	8.2	9.8		
		MCDBR	10.6	12.9	14.2	17.3	19.4	16.9	16.2	13.8	12.7	10.7	10.1	10.1		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.295	0.321	0.356	0.375	0.396	0.395	0.385	0.366	0.332	0.304	0.285		
	taud		2.368	2.362	2.372	2.332	2.371	2.346	2.371	2.397	2.436	2.502	2.515	2.453		
	Ebn at Noon		256	281	288	286	283	276	275	274	269	265	253	249		
	Edh at Noon		23	29	33	37	37	38	37	35	31	25	20	19		
(44) All-Sky Solar Radiation	RadAvg		371	660	1117	1455	1770	1911	1915	1621	1241	731	390	277		
	RadStd		42	72	110	148	157	146	89	99	106	81	51	27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page